
VVU EARTH TECH

Fabricator Specification Guide — Manufacturing Partnership Edition

FOR OUTREACH DISTRIBUTION, SCOPING & SALES FRAMEWORK STRATEGIES

Target Audience: Hardware Manufacturers, Fabrication Partners, Quality Engineers

CLASSIFICATION: Outreach Distribution — Scoping & Sales Framework Strategy

Content Scope: Technical, Operational, and Strategic — Excludes Legal & Financial

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1. Hydro-Gateway Hardware Overview

The **Hydro-Gateway** is the IoT sensor bridge that connects physical water infrastructure monitoring to the Epistemic DAG Runtime. It consists of an acoustic leak detection sensor array, a processing unit, and a communications module that transmits verified telemetry to the Epistemic Runtime via the AIR Kernel.

Hydro-Gateway Architecture

Component	Function	Interface	Evidence Flow
Acoustic Sensor Array	Detect water leak signatures via acoustic monitoring	Analog → DSP pipeline	Raw acoustic data → Epistemic Runtime as Fact
Processing Unit	Signal processing, pattern recognition, telemetry formatting	DSP → CAN/UART	Processed telemetry → canonicalized → SHA-256 hashed
Communications Module	Secure telemetry transmission to AIR Kernel	LTE/WiFi/NB-IoT	Signed telemetry → NATS durable queue → Acceptance Pipeline
Power Management	Battery + solar harvesting for continuous operation	Li-ion + solar panel	Power status → Fact Log (monitoring evidence)
Enclosure	IP67-rated environmental protection	Marine-grade aluminum	Deployment metadata → Fact Log

2. 24 Parametric Constraints

The Hydro-Gateway operates under 24 parametric constraints that define acceptable operational boundaries. These constraints are verified at fabrication (FAI), prototype testing, and field deployment stages.

Constraint	Specification	Category	Component
Acoustic sensitivity	≥ 45 dB SNR	DSP pipeline	Sensor array
Frequency range	20 Hz – 200 kHz	Signal processing	Acoustic sensor
Detection accuracy	≥ 95% (controlled)	Pattern recognition	Processing unit
False positive rate	≤ 5% (controlled)	Classification	Processing unit
Operating temperature	-10°C to +55°C	Environmental	Enclosure + electronics
Humidity tolerance	0–99% RH (non-condensing)	Environmental	Enclosure
Depth rating	IP67 (1m, 30min)	Ingress protection	Enclosure
Battery life	≥ 72 hours continuous	Power management	Battery + solar
Solar harvesting	≥ 15Wh/day (average SA conditions)	Power management	Solar panel
Communication latency	≤ 500ms (p99)	Communications	LTE/NB-IoT module
Data integrity	SHA-256 verified per packet	Cryptographic	Processing unit
Clock synchronization	HLC (wall_time, logical, node_id)	Temporal	Processing unit
Memory	≥ 512KB buffer	Storage	Processing unit
Processing throughput	≥ 50 telemetry events/sec	Performance	Processing unit
Mounting	Standard municipal pipe fittings	Mechanical	Enclosure mounting
Weight	≤ 2.5 kg (deployed unit)	Mechanical	Full assembly
Vibration resistance	IEC 60068-2-6	Mechanical	Enclosure + internals
EMC compliance	IEC 61000-6-2/6-4	Electromagnetic	Full assembly
Mean time between failures	≥ 10,000 hours	Reliability	Full assembly
Field life	≥ 10 years	Durability	Full assembly
Upgrade path	Hot-swappable sensor module	Maintenance	Modular design
Telemetry format	RFC 8785 JCS + SHA-256	Canonicalization	Processing unit
Signing capability	Ed25519 onboard	Cryptographic	Secure element
Kill switch compliance	Distributed kill switch integration	Safety	Processing unit

3. Acoustic Leak Detection Sensor Specifications

The acoustic sensor array is the primary sensing element of the Hydro-Gateway. It detects water leak signatures through continuous acoustic monitoring of pipe infrastructure, using pattern recognition to distinguish leak events from background noise.

Parameter	Specification	Measurement Method
Sensor type	Piezoelectric acoustic transducer	Component specification
Sensitivity	≥ 45 dB SNR at 1m distance	Laboratory calibration
Frequency response	20 Hz – 200 kHz (flat ±3 dB)	Frequency sweep test
Sampling rate	≥ 44.1 kHz	DSP pipeline configuration
Dynamic range	≥ 80 dB	Laboratory measurement
Directionality	Omnidirectional (pipe-mounted)	Field deployment configuration
Temperature compensation	Automatic (-10°C to +55°C)	Onboard calibration loop
Self-test	Daily acoustic calibration pulse	Automated test sequence

4. Materials & Fabrication Requirements

Component	Material	Specification	Fabrication Process
Enclosure body	Marine-grade aluminum (6061-T6)	IP67, anodized finish	CNC machining + anodization
Enclosure seals	Viton O-ring (fluoropolymer)	IP67 compliant, -20°C to +200°C	Precision molding
Sensor housing	316L stainless steel	Corrosion resistant, biocompatible	CNC machining + passivation
PCB substrate	FR-4 (multi-layer)	4-layer, 1.6mm, ENIG finish	Standard PCB fabrication
Solar panel frame	Anodized aluminum	UV-resistant, weather-proof	Extrusion + anodization
Mounting bracket	Galvanized steel	Standard municipal pipe fittings	Stamping + galvanization
Antenna enclosure	UV-resistant polycarbonate	IP54, impact resistant	Injection molding

5. Assembly Requirements

- **Assembly Sequence:** Defined step-by-step assembly procedure with quality gates at each stage.
- **Torque Specifications:** All fasteners torqued to specification with calibrated tools.
- **Seal Verification:** IP67 seal verification via pressure test after enclosure assembly.
- **PCB Installation:** PCB mounted with ESD-safe procedures; conformal coating applied to specified areas.
- **Sensor Calibration:** Each acoustic sensor individually calibrated against reference standard.
- **Communications Test:** Full communications module test (LTE/NB-IoT) before enclosure sealing.
- **Integration Test:** End-to-end telemetry flow test: acoustic event → processing → signing → transmission.
- **Final Inspection:** Visual inspection, dimensional verification, and functional test before shipment.

6. Quality Assurance Procedures

QA Stage	Procedure	Standard	Acceptance Criteria
Incoming Inspection	Material certification + dimensional check	ISO 9001	All materials meet specification
In-Process Inspection	Assembly step verification at each gate	Work instruction	Each gate PASS before next step
PCB Functional Test	Electrical test + firmware verification	IPC-6012	All test points within specification
Sensor Calibration	Acoustic sensitivity + frequency response	Laboratory standard	SNR $\geq$ 45 dB, frequency $\pm$ 3 dB
IP67 Verification	Pressure test (1m, 30min water immersion)	IEC 60529	No ingress detected
Communications Test	Network registration + data transmission	Operator specification	Successful telemetry transmission
Integration Test	End-to-end acoustic $\rightarrow$ telemetry flow	System specification	SHA-256 verified telemetry received
Final Inspection	Visual + dimensional + functional	Customer specification	All 24 parametric constraints met

7. Testing & Verification

First Article Inspection (FAI)

- **FAI Purpose:** Verify that the first production article meets all 24 parametric constraints.
- **FAI Scope:** Full dimensional, functional, and environmental test suite.
- **FAI Documentation:** Results recorded as Facts in the Epistemic Runtime — immutable evidence.
- **FAI Verification:** Independent verification against specification; SHA-256 signed results.

Production Quality Audit

- **Sampling Plan:** Statistical sampling per AQL (Acceptable Quality Level) framework.
- **Environmental Test:** Temperature cycling, humidity exposure, vibration per IEC 60068.
- **Reliability Test:** Accelerated life testing for MTBF verification.
- **Evidence Recording:** All test results recorded as Facts; audit trail immutable.

8. Prototype Development Lifecycle

Phase	Objective	Deliverable	Quality Gate
P0 — Concept	Define specifications, constraints, interfaces	Specification document	Review + approval
P1 — Design	Detailed design, material selection, fabrication planning	Design package (CAD, BOM, process)	Design review
P2 — Prototype Build	Fabricate first article, assemble, calibrate	Functional prototype	FAI (all 24 constraints)
P3 — Integration Test	Connect to Epistemic Runtime, verify telemetry flow	Integrated system test report	End-to-end telemetry verified
P4 — Field Trial	Deploy at Cape Town pilot site, monitor performance	Field trial data + analysis	Performance meets specification
P5 — Production Readiness	Finalize fabrication process, QA procedures, supply chain	Production readiness review	All gates PASS

9. Epistemic Runtime Integration

The Hydro-Gateway integrates with the Epistemic DAG Runtime through the AIR Kernel. Every telemetry event from the Hydro-Gateway follows this path:

- **1. Sensing:** Acoustic sensor detects event → DSP processing → telemetry packet.
- **2. Canonicalization:** Telemetry canonicalized via RFC 8785 JCS.
- **3. Hashing:** SHA-256(JCS(telemetry)) produces deterministic fingerprint.
- **4. Signing:** Ed25519 signature created on the Hydro-Gateway secure element.
- **5. Transmission:** Signed telemetry transmitted via LTE/NB-IoT to NATS durable queue.
- **6. Acceptance:** AIR Kernel Acceptance Pipeline processes telemetry through 5-Pass Evidence Compiler.
- **7. Evidence:** Accepted telemetry becomes a Fact in the Fact Log, indexed in MMR.
- **8. Decision:** Policy evaluation produces Projections and Decisions based on the Fact.

10. Cape Town Pilot Hardware Context

Cape Town serves as the alternative pilot municipality case study for the VVU EARTH TECH Hydro-Gateway deployment. Cape Town was selected following a governance review that identified operational risks at the originally proposed municipality. Cape Town's established water management infrastructure, progressive governance framework, and existing IoT deployment experience make it an ideal partner for validating municipal-grade infrastructure monitoring technology.

Hardware Deployment Requirements

- **Sensor Placement:** Hydro-Gateway units at designated monitoring points in Cape Town water network.
- **Communications:** LTE/NB-IoT connectivity for telemetry transmission.
- **Power:** Battery + solar harvesting; 72-hour minimum continuous operation.
- **Environmental:** IP67 rated for outdoor deployment in Cape Town climate conditions.
- **Quantity:** Initial pilot deployment scope to be determined in partnership discussions.

11. Partnership & Supply Chain Strategy

7-Track Resource Acquisition & Partnership Strategy: **Track A — Municipal Partnerships:** Active outreach to Cape Town and progressive municipalities for pilot deployment and validation partnerships. *Status: Active Outreach* **Track B — University Research Partnerships:** Engagement with South African and international universities for collaborative research, validation observation, and academic publication. *Status: Strategy* **Track C — Industry Integration:** Partnership development with water infrastructure, IoT, and municipal technology companies for integration, supply chain, and co-development. *Status: Strategy* **Track D — Public Funding:** Application to national and provincial technology innovation funds, water sector grants, and digital infrastructure programmes. *Status: Strategy* **Track E — Private Investment:** Structured engagement with impact investors, technology venture capital, and ESG-aligned funds. *Status: Strategy* **Track F — Sponsorship & Equipment:** Outreach to hardware manufacturers, cloud providers, and technology sponsors for equipment, infrastructure, and in-kind support. *Status: Strategy* **Track G — Community & Open Source:** Building developer community, open-source contributors, and civic technology networks for grassroots validation and adoption. *Status: Strategy*

Track F — Sponsorship & Equipment (Fabricator-Relevant)

- **Target Partners:** PCB fabricators, sensor manufacturers, enclosure fabricators, communications module suppliers.
- **Requested Support:** Prototype fabrication, component supply, FAI testing support.
- **Status:** Strategy — outreach planned, not yet active.